

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

INFR09027 COMPUTER COMMUNICATIONS AND NETWORKS

Wednesday 10th May 2017

14:30 to 16:30

INSTRUCTIONS TO CANDIDATES

Answer any TWO of the three questions. If more than two questions are answered, only QUESTION 1 and QUESTION 2 will be marked.

All questions carry equal weight.

CALCULATORS MAY NOT BE USED IN THIS EXAMINATION

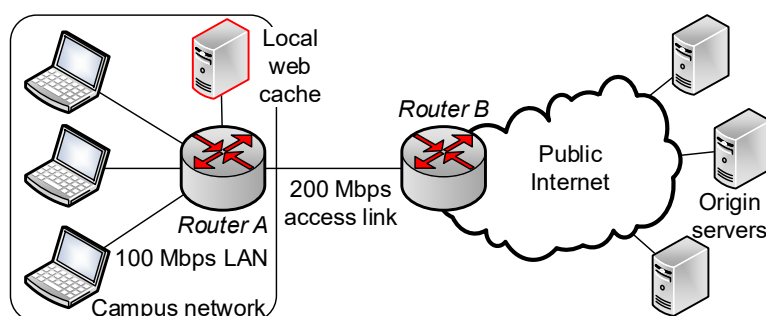
Year 3 Courses

Convener: C. Stirling

External Examiners: A. Cohn, A. Donaldson, S. Kalvala

THIS EXAMINATION WILL BE MARKED ANONYMOUSLY

1. (a) Name four properties of packet switching and discuss each of them briefly. [4 marks]
- (b) Suppose that a link has 1 Gbps capacity and 100ms round-trip time. What is the amount of data that needs to be put on the link every 100ms in order to efficiently utilise the link? If the amount is 10 Mbits every 100ms, what is link utilisation? Given the computed link utilisation, briefly explain whether the link is efficiently utilised or not. [5 marks]
- (c) Consider a campus network environment below. An access link capacity is 200 Mbps and the link capacity inside the campus network is 100 Mbps.



Users in the campus network request web objects to origin servers via a local web cache. If the web cache has the requested objects, it directly provides them to the users. Otherwise, the web cache retrieves those web objects from origin servers and returns them to the users while caching them at the same time. Assume that the average object size is 100 Kbits and 900 requests per second are generated on average from the users.

Suppose that round-trip propagation delay between router A and any origin server is 50ms, and round-trip propagation delay in the campus network is zero. In addition, suppose that for two links (one from router B to router A and the other from router A to the web cache), a queuing delay is 10ms if link utilisation is less than 50%, and 100ms otherwise. For other remaining links, assume that queuing delay is zero. Ignore processing and transmission delays. Is it useful to run the local web cache if the cache hit rate is 40%? Justify your answer.

[8 marks]

QUESTION CONTINUES ON NEXT PAGE

QUESTION CONTINUED FROM PREVIOUS PAGE

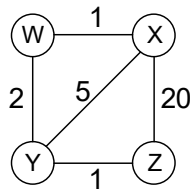
- (d) Consider a problem of distributing a file of size F from one server to N peers under the Peer-to-Peer (P2P) model that was studied in the course. Let D_{p2p} be the time to distribute the file to N peers, which meets the following inequality:

$$D_{p2p} \geq \max \left\{ \frac{F}{u_s}, \frac{F}{d_{min}}, \frac{NF}{u_s + \sum_{i=1}^N u_i} \right\}.$$

Note that u_s is the upload capacity of server s , d_{min} is the download rate of the slowest peer, and u_i is the upload capacity of peer i . Suppose that all peers have the same upload capacity u (i.e., for any i , $u_i = u$), $u_s = 100u$, and $d_{min} = 11u$. What is the minimum number of peers (N) that makes D_{p2p} governed not by the smallest download rate (d_{min}) but by the upload capacity of server and peers? In other words, as the number of peers increases, d_{min} has no effect on D_{p2p} . Find a minimum N such that d_{min} is no longer the dominant factor of D_{p2p} . Discuss the implication of the above case where $u_s \gg d_{min}$.

[8 marks]

2. (a) Assume we have a go-back-N (GBN) receiver. On receipt of a packet that is in-order, all it does is deliver the data to the application layer. In all cases, it acknowledges the packet that is the last in-order packet that has arrived. Suppose a host that runs a go-back-N receiver wants to communicate with a TCP sender. Would that work? In other words, is reliable data transfer possible? If necessary, additional assumptions can be made. Justify your answer. [4 marks]
- (b) A selective-repeat receiver selectively acknowledges the packets that it receives. Now suppose that there is a host that runs a go-back-N receiver and a selective-repeat sender. Would that work? Justify your answer. [4 marks]
- (c) Consider a network which has a lot of bursty losses (several packets lost in a burst) on the wire. Which of the two protocols functions better in such scenarios—go-back-N or TCP? Justify your answer. [5 marks]
- (d) Consider the following network consisting of 4 nodes. An integer on an edge denotes communication cost between two nodes.



Using the distance vector algorithm, complete the following routing tables. To get you started, initial routing tables of round 0 are shown below. [12 marks]

QUESTION CONTINUES ON NEXT PAGE

QUESTION CONTINUED FROM PREVIOUS PAGE

Round 0:

W table					X table					Y table					Z table								
		cost to						cost to						cost to						cost to			
		w	x	y	z			w	x	y	z			w	x	y	z			w	x	y	z
from	w	0	1	2	∞	from	w	∞	∞	∞	∞	from	w	∞	∞	∞	∞	from	w	∞	∞	∞	∞
	x	∞	∞	∞	∞		x	1	0	5	20		x	∞	∞	∞	∞		x	∞	∞	∞	∞
	y	∞	∞	∞	∞		y	∞	∞	∞	∞		y	2	5	0	1		y	∞	∞	∞	∞
	z	∞	∞	∞	∞		z	∞	∞	∞	∞		z	∞	∞	∞	∞		z	∞	20	1	0

Round 1:

from		cost to				from		cost to				from		cost to				from		cost to			
		w	x	y	z			w	x	y	z			w	x	y	z			w	x	y	z
	w						w						w						w				
	x						x						x						x				
	y						y						y						y				
	z						z						z						z				

Round 2:

from		cost to				from		cost to				from		cost to				from		cost to			
		w	x	y	z			w	x	y	z			w	x	y	z			w	x	y	z
	w						w						w						w				
	x						x						x						x				
	y						y						y						y				
	z						z						z						z				

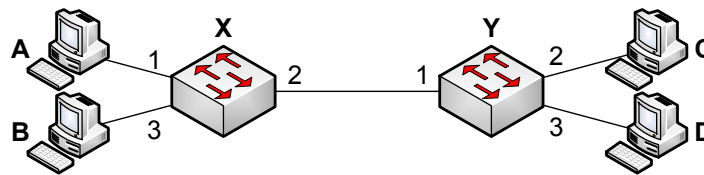
Round 3:

from		cost to				from		cost to				from		cost to				from		cost to			
		w	x	y	z			w	x	y	z			w	x	y	z			w	x	y	z
	w						w						w						w				
	x						x						x						x				
	y						y						y						y				
	z						z						z						z				

3. (a) Discuss two differences between a router and an Ethernet switch. [4 marks]

(b) Consider a network where there are only two nodes that try to send one frame to each other. Suppose that the two nodes detect collision at the same time. Now recall the CSMA/CD algorithm. Upon collision detection, the two nodes conduct binary exponential backoff as follows. After m^{th} collision, the nodes choose K at random from $\{0, 1, 2, \dots, 2^m - 1\}$ and wait for K time units. Afterwards the nodes retransmit their frame when the channel is idle. Let $P_b(t=m)$ be the probability that the two nodes successfully send their frame after m^{th} collision (i.e., on the $(m + 1)^{th}$ attempt of transmission). Assume that $P_b(t=0)$ is 0.5. Calculate $P_b(t=1)$ and $P_b(t=2)$. Based on the computed probabilities, discuss the effectiveness of the binary exponential backoff mechanism. Note that probability can be expressed as a fraction and there is no need to convert a fraction into a decimal. [6 marks]

(c) The following questions are about the self-learning mechanism in a switch. Suppose that all of the hosts (A , B , C and D) in the diagram below have a fully-populated ARP table. This means that each host knows the up-to-date mapping between the IP address and the MAC address for all hosts. Therefore, a host can immediately create an Ethernet frame and transmit it to the network.



- i. Briefly explain under what condition(s) a self-learning switch does forwarding, flooding and dropping of a frame. [6 marks]
- ii. Consider the above Ethernet network. Initially, switches X and Y have no entry in their switch table. Note that an entry in the switch table consists of two fields: MAC address and interface number. From the above figure, a number denotes an interface number, and MAC addresses and host names are denoted as A , B , C and D . Host A sends a frame to host B which in turn sends a reply frame back to host A . After one round of frame exchange to each other, what entry (entries) do switches X and Y contain in their switch table? List the switch table entry (entries) for each switch and briefly discuss why each switch has it (them). [4 marks]
- iii. Given the table entry (entries) at switches X and Y in part ii., suppose that switch X is accidentally rebooted. Further assume that the entry (or entries) in switch Y has/have not expired. Now host B sends a new frame to host A once the switch reboot is completed. Given the situation, discuss what action(s) switch Y would take. [5 marks]